MD ABU BAKAR

Python Developer | Flask Backend Developer | Full Stack Web Development | Final-Year CSE Student
Uttar Dinajpur(W.B), India | GitHub: github.com/MdAbuBakar209 | LinkedIn: linkedin.com/in/md-abu-bakar

PROFESSIONAL SUMMARY

Final-year Computer Science student with hands-on experience in Python, Flask, SQLite, and backend web development. Built and deployed real-world applications including a Student Management System with authentication and CRUD functionality. Also working on a Deep Reinforcement Learning research project focused on IoT resource allocation. Passionate about backend engineering, problem-solving, and building scalable applications.

TECHNICAL SKILLS

Languages:	Python, Java, JavaScript, HTML, CSS
Frameworks:	Flask, Bootstrap
Databases:	SQLite
Libraries:	PyTorch, NumPy, Matplotlib
Tools:	Git, GitHub, Render, Netlify, GitHub Codespaces
CS Fundamentals:	Data Structures & Algorithms | Object-Oriented Programming (OOP)

PROJECTS

Student Management System —Full Stack Web App | GitHub: github.com/MdAbuBakar209/student-management-system-flask

• Developed a full-stack Flask web application with user authentication and session management.
• Implemented CRUD operations for managing student records with responsive dashboard UI.
• Integrated SQLite database, flash messaging, search functionality, and confirmation alerts.
• Deployed application on Render and maintained source code on GitHub.

Task Manager CLI Application | GitHub Repo: github.com/MdAbuBakar209/task-manager-python

• Built a command-line task manager with add, update, delete, and status tracking functionality.
• Implemented persistent task storage using JSON file handling.
• Demonstrated project using GitHub Codespaces.

Personal Portfolio Website | GitHub Repo: github.com/MdAbuBakar209/portfolio-website
HTML | CSS | Netlify

• Designed and deployed a responsive personal portfolio website showcasing projects and skills.

Edge IoT Resource Allocation : — College Research Project (In Progress)

• Developing a Deep Reinforcement Learning framework for intelligent resource allocation in MEC-enabled IoT systems.
• Implementing Double DQN and Dueling Network architectures using PyTorch.
• Applying Lyapunov Optimization techniques for long-term queue stability and energy efficiency.

EDUCATION

B.Tech in Computer Science & Engineering — Oct 2022 – July 2026
Ramkrishna Mahato Government Engineering College, Purulia, West Bengal

Higher Secondary (XII) — Pure Science (PCM) — Apr 2019 – Mar 2020
Jawahar Navodaya Vidyalaya | Grade: A+ | 82.4%

Secondary (X) — Jan 2017 – Mar 2018
Manora High School (H.S) | Grade: A+ | 86.85%

LANGUAGES : **English** (Professional Working Proficiency) | **Hindi** (Native) | **Bengali** (Native) | **Urdu** (Native)